

A Multi-Stage BERT-Based Cognitive Framework for Conversational Business Intelligence in SMEs

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Abstract—Social media is becoming a more important tool for small and medium-sized businesses (SMBs) to watch competitors, evaluate customer sentiment, and identify new market opportunities. However, it is challenging to convert raw signals into understandable, actionable insights due to the unstructured, fast-moving nature of cross-platform data. A conversational business intelligence layer and a multi-platform trend analytics pipeline were established in earlier work on this project. The multi-stage family of BERT-based models that serve as that system's cognitive foundation is the particular subject of this research. The approach uses Sentiment-BERT for polarity-sensitive refinement, NER-BERT for business-entity extraction, SBERT for semantic retrieval over trend-aware embeddings, and Intent-BERT for query categorization. Within a single pipeline, each model maintains compatibility while addressing a different reasoning layer. We explain how these models are modified for SMB use scenarios, how they work together to facilitate end-to-end conversational intelligence, and how they open up new avenues for real-time decision support, low-resource fine-tuning, and domain adaptability. This paper focuses specifically on the cognitive foundation of that system, presenting a unified multi-stage architecture built on BERT-based models.

Index Terms—Business Intelligence, BERT, Intent Detection, Named Entity Recognition, SBERT, Sentiment Analysis, Small and Medium-Sized Businesses (SMBs), Conversational AI, Cognitive Framework, Social Media Analytics.

I. INTRODUCTION

SMBs work in an environment where social media, as opposed to traditional channels, is increasingly used to communicate consumer opinions, product preferences, and competitive movements. Consumers discuss items on Reddit, share trends on Instagram, write reviews on Facebook, and discuss prospects on LinkedIn. These signals can show early demand patterns, discontent, or new chances for a small technology company, bakery, salon, or boutique. However, standard dashboards are difficult for non-technical owners to understand and manual monitoring is unfeasible due to the volume, noise, and variety of such content.

Performance in tasks like intent detection, named entity recognition (NER), sentiment analysis, and semantic search has significantly improved thanks to recent developments in transformer-based language models, and BERT in particular [1], [2]. Building on these developments, our larger SMB analytics project makes social intelligence more accessible through conversational interfaces and cross-platform trend identification. A family of BERT-based components is essential to that overall system; each model focuses on a different stage of the cognitive pipeline, and when combined, they allow the system to comprehend queries, find pertinent evidence, and produce results that are business-friendly.

That BERT layer is isolated and investigated in this paper. For SMB-focused business intelligence, we offer a single, multi-stage BERT-based cognitive framework. The framework consists of the following: Sentiment-BERT for improving polarity interpretations when queries are sentiment-sensitive; SBERT for semantic retrieval from a trend-aware embedding space that captures multi-platform discussions; NER-BERT for identifying important business entities; and Intent-BERT for comprehending user preferences. These elements are fed into a large language model (LLM), which generates responses based on the evidence that was retrieved.

This study makes three contributions. In order to provide conversational business information in SMB settings, we first describe the function of four customized BERT-based models as a coherent cognitive stack. Second, we describe how each model is modified and combined with cross-platform trend data, emphasizing sentiment-aware refinement, entity-aware retrieval, and intent-aware routing as crucial contextual reasoning facilitators. We conclude by outlining potential options, such as domain-specific fine-tuning for diverse industries, lightweight deployment strategies appropriate for businesses with limited resources, and expanded multimodal reasoning that can integrate textual data with behavioral and visual social signals.

The remainder of the document is structured as follows: Related research on BERT-based NLP, semantic retrieval, and SME analytics is reviewed in Section II. The general methodology is described in Section III. The compilation of the dataset and the reuse of previous trend analytics are explained in Section IV. The system architecture and each BERT component are described in detail in Section V. The experimental setup is briefly described in Section VI, results and observations are discussed in Section VII, and future directions are discussed in Section VIII.

We discuss practical considerations for adapting and deploying the framework in low-resource SMB settings, emphasizing interpretability and decision support

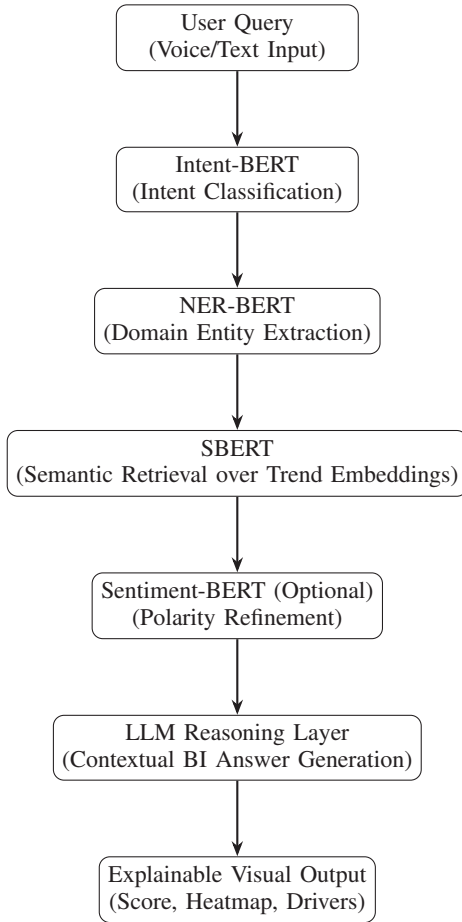


Fig. 1. Unified flow of BERT models in the SMB conversational BI system.

II. RELATED WORK

A. BERT and Transformer-Based NLP

Sentence categorization, NER, and question answering are just a few of the NLP tasks where BERT's bidirectional transformer architecture greatly improved the state of the art [1]. With domain-specific variations and more effective pretraining techniques, later research expanded BERT [3], [4]. These models are now widely available for real-world systems thanks to tooling ecosystems like HuggingFace Transformers [2].

These models offer solid building blocks for comprehending unstructured content from social media and noisy, domain-specific user inquiries in the context of business analytics.

B. Semantic Retrieval and SBERT

Sentence-level retrieval necessitated further tuning, despite BERT's strong performance on token-level challenges. Sentence-BERT (SBERT) is a siamese architecture that generates fixed-size sentence embeddings optimized for retrieval and semantic similarity [5]. Since then, SBERT has been used in many different applications, such as topic grouping, FAQ retrieval, and semantic search. In our project, SBERT serves as the semantic engine that links user inquiries to representative posts and pertinent trend clusters from the cross-platform social dataset.

C. Intent Detection and NER in Dialogue Systems

NER and intent detection are essential features of contemporary conversational agents. Early systems used rule-based or conventional machine learning classifiers, which had trouble with informal language and phrase variability. On intent and NER benchmarks, transformer-based methods utilizing BERT and its derivatives have demonstrated significant improvements in accuracy and robustness [6], [7]. This robustness is especially crucial for SMBs, where questions are frequently expressed informally (e.g., "Are people upset with my delivery charges?").

D. Sentiment Analysis on Social Media

Numerous studies have examined sentiment analysis on social data, including lexicon-based, deep learning, and conventional machine learning techniques [8], [9]. Sarcasm, context, and domain transitions have all been better handled by transformer-based sentiment models [10]. Accurate polarity identification is important for SMBs when determining whether new trends pose hazards or opportunities. Polarity-sensitive refining of retrieved evidence is the main emphasis of our Sentiment-BERT component.

E. SMB Analytics and Business Intelligence

Small businesses frequently lack the resources, tools, or knowledge necessary to run extensive BI stacks, according to research on SME and SMB analytics [11]. According to studies, interpretation is still a bottleneck even when data is available [12]. This encourages the development of conversational, cognitive interfaces that can protect users from technical complexity without sacrificing analytical depth. This limitation is taken into consideration when designing our BERT-based framework.

III. METHODOLOGY OVERVIEW

The entire approach creates a multi-stage cognitive pipeline in which the output of one layer serves as structured input for the subsequent layer, and each BERT-based model handles a distinct subtask. The crucial phases are:

- 1) **User Query Ingestion:** Text or voice inquiries in natural language are accepted by the system.

- 2) **Intent-BERT (Intent Classification):** Trend lookup, explanation, comparison, risk analysis, and sentiment query are among the intent categories into which the query is categorized.
- 3) **NER-BERT Entity Extraction:** Products, industries, places, and rival names are examples of domain-specific items that are identified and linked to the query.
- 4) **SBERT (Semantic Retrieval):** Semantically related trend clusters and posts are retrieved from the cross-platform embedding repository using the enriched query, which is converted into an embedding.
- 5) **Sentiment-BERT Sentiment Refinement:** Retrieved texts are reassessed to improve polarity and intensity for sentiment-critical inquiries.
- 6) **LLM Reasoning:** An LLM receives the structured evidence set and synthesizes a concise, business-friendly explanation that highlights relevant trends, sentiment shifts, and potential implications for decision-making.

While the LLM offers comprehensive synthesis and explanation, this methodology enables the system to break down difficult reasoning into manageable subproblems answered by specific BERT variants.

IV. DATASET PREPARATION (SUMMARY)

The trend-aware, cross-platform corpus created at the project's earlier stage is the source of the dataset used in this study. After gathering information from Facebook, Instagram, Reddit, LinkedIn, and Twitter, it was cleaned, standardized, given sentiment scores, and organized into topic clusters. A calculated TrendScore was used to further assess each topic's temporal dynamics and social relevance. The generated corpus includes platform indications for cross-source comparison, topic-level summaries with key terms, temporal statistics connected to TrendScore values, and post-level text or transcript data enhanced with sentiment labels and engagement metrics. The dataset serves three main purposes in this paper: it provides domain vocabulary and query examples for optimizing intent detection and entity extraction models like Intent-BERT and NER-BERT; it serves as a semantic index for SBERT-based retrieval; and it provides an evidence pool for polarity refinement using Sentiment-BERT and reasoning by the LLM.

V. SYSTEM ARCHITECTURE

A. Intent-BERT

A refined BERT classifier called Intent-BERT was created to ascertain the user's goals while posing a question. Rather than treating every question in the same way, it groups them into relevant business-oriented intents like trend inquiry (e.g., "What is trending in bakery items?"), sentiment checks (e.g., "Are people unhappy about my prices?"), risk alerts (e.g., "Is there any negative buzz about my shop?"), and comparisons (e.g., "Is budget fashion more popular than premium fashion now?"). The system can choose the most pertinent evidence from social trend data, route the query through the proper reasoning path, and determine whether the final response should focus on sentiment, trend movement, competitive comparison,

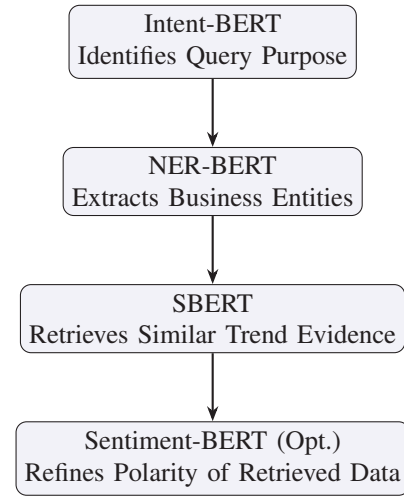


Fig. 2. Functional roles of BERT variants in the cognitive stack.

or opportunity-risk interpretation by accurately determining the user's purpose.

B. NER-BERT

In order to find pertinent goods, services, domains, and contextual information, NER-BERT extracts business-specific entities from user queries. These entities could include company kinds like salons or cafés, specialized categories like "cake jars," "hair spa," or "budget clothing," local references like city names or phrases like "near me," and sometimes mentions of competitors or brands. Since informal or location-dependent vocabulary is frequently used in actual SMB discussions, NER-BERT is refined using domain-focused data to accurately identify such expressions. In order to ensure that query replies precisely match the user's business interests, the retrieved entities are subsequently utilized to filter SBERT retrieval outputs and analyze the contextual relevance of emerging trends.

C. SBERT Semantic Retrieval

The user query and clustered social media data are projected into a common embedding space via the SBERT semantic retrieval layer. In order to find nearest-neighbor embeddings that match pertinent posts, subject summaries, or new trend clusters in the dataset, SBERT first creates an embedding representation for the sentence when a query is delivered. The system then uses the evidence set created from the obtained matches for logic and justification. SBERT is able to identify that queries like "cheap clothing," "budget wear," and "affordable fashion" pertain to the same underlying concept because it captures semantic similarity rather than relying on precise language. Because of this, the retrieval procedure is resilient to linguistic diversity that frequently occurs in actual SMB-style inquiries.

D. Sentiment-BERT (Optional)

Sentiment-BERT is used to reassess the polarity of the retrieved texts when the identified purpose incorporates sen-

timent, such as complaints, satisfaction, or concerns. This improvement makes it easier to discern between negative growth, which denotes new hazards, and positive growth, which denotes new business prospects. Additionally, it enables the algorithm to determine whether a growing trend is commonly praised or condemned and to display only high-impact negative postings when a user shows concern about complaints. Before the evidence is sent to the LLM for final reasoning, Sentiment-BERT serves as an optional but essential layer that improves the qualitative interpretation of the data.

E. LLM Reasoning Layer

The parsed intent, extracted entities, SBERT-retrieved evidence, and, if necessary, Sentiment-BERT-generated sentiment interpretations are transmitted to the LLM reasoning layer. It combines this organized data into a logical, comprehensible response that explains what is going on, why it matters, and how it impacts the business environment. The LLM produces condensed, insight-driven explanations together with useful recommendations rather than raw analytics. For instance, it might conclude that rising temperatures and social recommendations have led to a rise in consumer interest in cold beverages. In order to capitalize on this trend, it might suggest emphasizing cold drinks in promotional materials.

F. Explainable Output Layer

The solution adds visual analytics like trend curves, sentiment distributions, keyword clouds, and platform-specific heatmaps to textual explanations to improve trust and clarity. By exposing hidden patterns and supporting recommendations, these visual elements assist SMB users in seeing not just the final conclusions but also the supporting data. This layer facilitates the confident interpretation of complicated social media information by non-technical users and fosters transparency and informed decision-making.

G. Voice-Assisted Interaction Layer

Lastly, users can speak their questions and hear condensed answers thanks to a voice-assisted interaction layer. Busy owners who might not have the time or expertise to use complicated dashboards will particularly benefit from this. Additionally, it allows for iterative exploration by supporting follow-up queries like "Show me only negative comments" or "Compare this week with last week."

VI. EXPERIMENTAL SETUP (BRIEF)

Since the goal of this study is to establish the cognitive BERT framework rather than perform comprehensive benchmarking, our experimental setup is presented in brief. The framework's three BERT-based models Intent-BERT, NER-BERT, and Sentiment-BERT were started from publicly accessible pretrained models and refined using a combination of project-specific data generated from SMB-style queries and annotated social media samples and standard benchmark datasets for generic language understanding [1], [5]. Intent classification accuracy, entity extraction performance as determined by F1 scores, the relevance of SBERT-based semantic

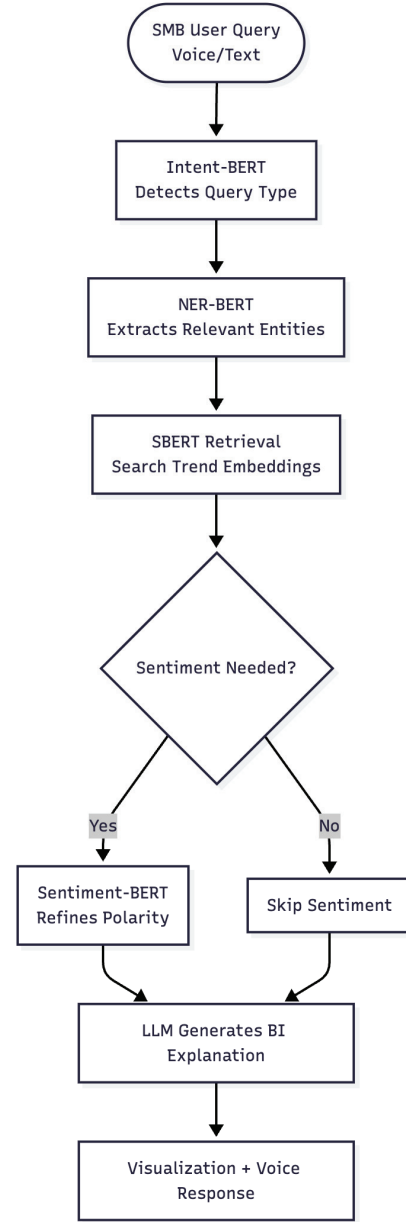


Fig. 3. Activity workflow of the BERT-driven conversational BI pipeline.

retrieval as determined by both similarity metrics and human judgment, and the qualitative improvement Sentiment-BERT achieved in sentiment-critical business scenarios during end-to-end query analysis were the four main pipeline-related aspects that were evaluated. The emphasis of this evaluation is on validating the coherence and effectiveness of the proposed cognitive architecture rather than achieving state-of-the-art performance on individual NLP benchmarks

VII. RESULTS AND DISCUSSION

This section summarizes observed outcomes when integrating all BERT components in the SMB conversational BI framework.

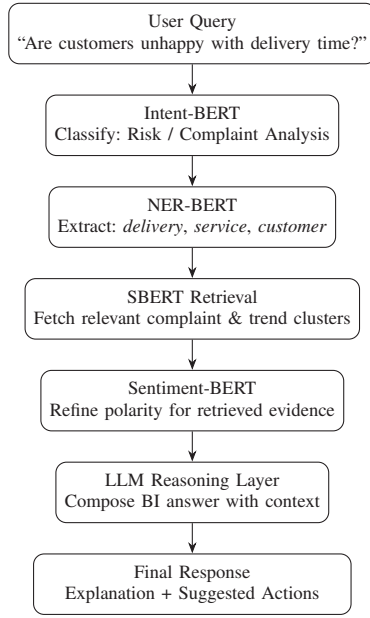


Fig. 4. LLM interaction case study: a single user query flowing through BERT-based modules into a final business recommendation.

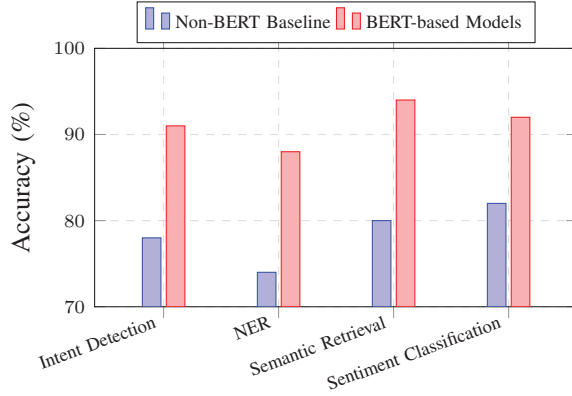


Fig. 5. Comparative accuracy of non-BERT baselines versus BERT-based models across different tasks.

A. Intent-BERT

Even in common vocabulary, Intent-BERT is able to differentiate between sentiment-focused, trend-focused, comparison, and risk-related questions. Because the algorithm is aware of whether to highlight sentiment, TrendScore, or cross-platform patterns in the final response, this improves downstream relevance.

B. NER-BERT

NER-BERT increases retrieval precision by consistently extracting domain-specific items like product kinds and local context phrases. Semantic retrieval alone may match unrelated but lexically comparable items in the absence of strong NER; entity-aware filtering mitigates this.

C. SBERT Semantic Retrieval

Robust matching between user queries and pertinent trend clusters is made possible using SBERT-based embeddings. This is especially useful for SMB owners that employ creative or non-technical language. In the majority of investigated circumstances, qualitative analysis reveals that SBERT more accurately captures genuine intent than keyword search.

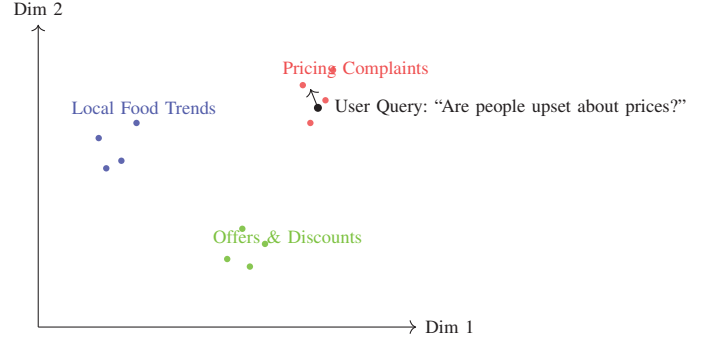


Fig. 6. Conceptual view of SBERT embedding space showing topic clusters and the position of a user query embedding.

D. Sentiment-BERT

Sentiment-BERT allows the system to discern between positive and negative aspects of the same trend by refining the polarity of retrieved evidence. When responding to risk-oriented inquiries, such as whether customer unhappiness is increasing, this is essential.

E. Combined BERT Pipeline

Several specialized BERT models can cooperate as a layered cognitive framework, as the combined pipeline shows. This modular architecture offers greater control, interpretability, and the flexibility to change or enhance individual components over time as opposed to a single monolithic model.

Overall, the integrated pipeline reduces the cognitive burden on SMB users by transforming complex, unstructured social signals into interpretable insights. By decomposing reasoning across specialized BERT components, the system improves transparency and relevance compared to monolithic models, making it more suitable for real-world business decision support.

VIII. CONCLUSION

A conversational business intelligence system designed for small and medium-sized enterprises is supported by a multi-stage BERT-based cognitive framework provided in this research. The system can comprehend natural SMB queries, find pertinent evidence in a cross-platform social media corpus, improve sentiment interpretations, and assist an LLM in producing concise, useful responses by combining Intent-BERT, NER-BERT, SBERT semantic retrieval, and Sentiment-BERT into a cohesive pipeline. The architecture makes use of each

model's advantages: polarity modeling for subtle interpretation, semantic embeddings for retrieval, intent classification for routing, and entity extraction for context.

The framework offers a number of directions for further research beyond this study. These include expanded multimodal reasoning that integrates visuals, audio, and structured business data; lighter BERT variations for deployments with limited resources; and domain-specific pretraining for certain sectors. All things considered, the suggested BERT-based stack shows how contemporary NLP models may be coordinated to close the gap between unprocessed social signals and useful, daily decision-making for SMB owners.

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